The Chinese University of Hong Kong (Shenzhen)
Advanced Math for AI
Fall 2026

Assignment 1

Issued: Friday 11th September, 2026 **Due:** Sunday 20th September, 2026

1. Solvability of Linear Regression **(40 points)**

Consider linear regression with two input features. Each training sample can be written as $\mathbf{x}^{(i)} = [x_1^{(i)}, x_2^{(i)}]^\top \in \mathbb{R}^2$ with scalar output $y^{(i)}$. The prediction function is

$$\widehat{y}^{(i)} = \mathbf{w}^\top \mathbf{x}^{(i)} + \beta = w_1 x_1^{(i)} + w_2 x_2^{(i)} + \beta,$$

where $\mathbf{w} = [w_1, w_2]^\top$ contains the regression weights and β is the intercept. For an exact fit, the prediction must satisfy $\widehat{y}^{(i)} = y^{(i)}$ for every training sample.

For three training samples, define the design matrix A, parameter vector $\boldsymbol{\theta}$, and target vector $\mathbf{y}$ by

$$A = \begin{bmatrix} x_1^{(1)} & x_2^{(1)} & 1 \\ x_1^{(2)} & x_2^{(2)} & 1 \\ x_1^{(3)} & x_2^{(3)} & 1 \end{bmatrix}, \qquad \boldsymbol{\theta} = \begin{bmatrix} w_1 \\ w_2 \\ \beta \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} y^{(1)} \\ y^{(2)} \\ y^{(3)} \end{bmatrix}.$$

The exact-fitting problem is therefore the linear system $A\boldsymbol{\theta} = \mathbf{y}$. It may have no solution, exactly one solution, or infinitely many solutions. In geometric terms, the regression model $y = w_1 x_1 + w_2 x_2 + \beta$ is a plane in (x_1, x_2, y)-space, and exact fitting asks for a plane that contains all three training points.

Consider the following three regression data sets.

Data set	$(\mathbf{x}^{(1)}, y^{(1)})$	$(\mathbf{x}^{(2)}, y^{(2)})$	$(\mathbf{x}^{(3)}, y^{(3)})$
1	$((0,0),1)$	$((1,0),3)$	$((0,1),4)$
2	$((0,0),1)$	$((1,0),2)$	$((2,0),4)$
3	$((0,0),1)$	$((1,0),2)$	$((2,0),3)$

(a) (18 points) For each of the three data sets:

i. Construct the design matrix A and target vector $\mathbf{y}$.

ii. Use Gaussian elimination to reduce the augmented matrix $[A \mid \mathbf{y}]$ to row-echelon or reduced row-echelon form. From the reduced system, determine whether $A\boldsymbol{\theta} = \mathbf{y}$ has no solution, exactly one solution, or infinitely many solutions.

iii. If solution(s) exists, find its complete set of solutions. (For infinitely many solutions, find the basis for the solution set.)

(b) (12 points) Recall that the null space of an $(m \times n)$ matrix A is defined by

$$\mathrm{Null}(A) = \{\mathbf{z} \in \mathbb{R}^n : A\mathbf{z} = \mathbf{0}\}.$$

For the design matrix A of each data set in Part (a):

i. Find Null(A) by giving a basis for it. Recall that the basis of the zero subspace is the empty set.

ii. Explain how the null space is related to uniqueness of the exact regression parameters.

(c) (10 points) When $A\boldsymbol{\theta} = \mathbf{y}$ has infinitely many solutions, let $\boldsymbol{\theta}_0$ be one particular solution. Show that every exact solution has the form

$$\boldsymbol{\theta} = \boldsymbol{\theta}_0 + \mathbf{z}, \qquad \mathbf{z} \in \text{Null}(A).$$

Hint: Show that $\boldsymbol{\theta} - \boldsymbol{\theta}_0 \in \text{Null}(A)$.

Solution:

(a) **Data set 1.**

$$[A \mid \mathbf{y}] = \left[\begin{array}{ccc|c} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 3 \\ 0 & 1 & 1 & 4 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Every column is a pivot column, so the system has exactly one solution, $\boldsymbol{\theta} = [2,\, 3,\, 1]^\top$, i.e. $w_1 = 2$, $w_2 = 3$, $\beta = 1$.

Data set 2.

$$[A \mid \mathbf{y}] = \left[\begin{array}{ccc|c} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 2 \\ 2 & 0 & 1 & 4 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

The last row reads $0 = 1$, so the system is inconsistent: there is no solution.

Data set 3.

$$[A \mid \mathbf{y}] = \left[\begin{array}{ccc|c} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 2 \\ 2 & 0 & 1 & 3 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Columns 1 and 3 are pivot columns and column 2 is free, so there are infinitely many solutions:

$$\boldsymbol{\theta} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + t \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \qquad t \in \mathbb{R},$$

i.e. $w_1 = 1$, $\beta = 1$, and w_2 arbitrary. A basis for the solution set (the homogeneous part) is $\{[0, 1, 0]^\top\}$.

(b) i. For data set 1, A has three pivot columns and is invertible, so $\text{Null}(A) = \{\mathbf{0}\}$ and its basis is the empty set $\emptyset$.

For data sets 2 and 3 the design matrix is the same, $A = \left[\begin{smallmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 2 & 0 & 1 \end{smallmatrix}\right]$, whose second column is zero, with z_2 free, so

$$\text{Null}(A) = \text{span}\left\{[0, 1, 0]^\top\right\},$$

with basis $\{[0, 1, 0]^\top\}$.

ii. Among the three data sets, only data set 1 has a unique exact solution, and this is precisely because $\text{Null}(A) = \{\mathbf{0}\}$ for its design matrix.

(c) ($\Leftarrow$) If $\mathbf{z} \in \text{Null}(A)$, then $A(\boldsymbol{\theta}_0 + \mathbf{z}) = A\boldsymbol{\theta}_0 + A\mathbf{z} = \mathbf{y} + \mathbf{0} = \mathbf{y}$, so $\boldsymbol{\theta}_0 + \mathbf{z}$ is an exact solution.

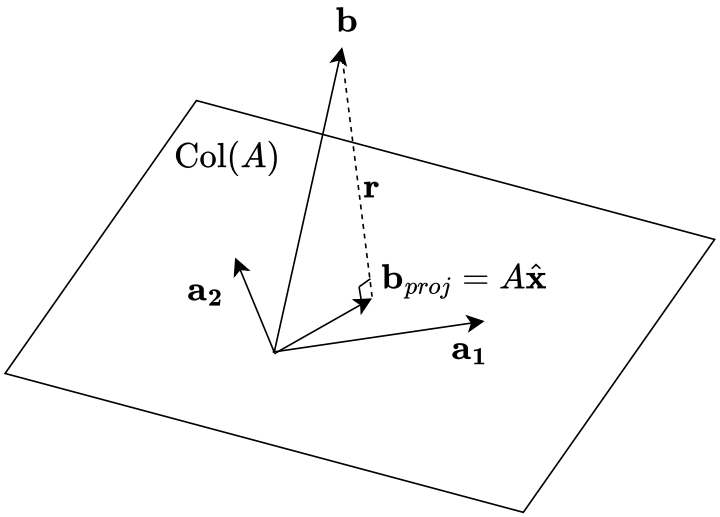
($\Rightarrow$) Conversely, let $\boldsymbol{\theta}$ be any exact solution and set $\mathbf{z} = \boldsymbol{\theta} - \boldsymbol{\theta}_0$. By linearity,

$$A\mathbf{z} = A\boldsymbol{\theta} - A\boldsymbol{\theta}_0 = \mathbf{y} - \mathbf{y} = \mathbf{0},$$

so $\mathbf{z} \in \text{Null}(A)$ and $\boldsymbol{\theta} = \boldsymbol{\theta}_0 + \mathbf{z}$. Hence the solution set is exactly $\boldsymbol{\theta}_0 + \text{Null}(A)$.

2. The Pseudoinverse **(25 points)**

If A is a square invertible matrix, then the system $A\mathbf{x} = \mathbf{b}$ has the unique solution $\mathbf{x} = A^{-1}\mathbf{b}$. For a rectangular matrix, however, an ordinary inverse does not exist. To obtain the closest approximate solution, we project $\mathbf{b}$ onto $\text{Col}(A)$ and choose $\widehat{\mathbf{x}}$ such that $A\widehat{\mathbf{x}}$ equals this projection.



Assume first that $A \in \mathbb{R}^{m \times n}$ has linearly independent columns, so that $A^\top A$ is invertible.

(a) (15 points) Show geometrically that

$$\widehat{\mathbf{x}} = (A^\top A)^{-1} A^\top \mathbf{b}.$$

Here $A^+ = (A^\top A)^{-1} A^\top$ is the Moore–Penrose pseudoinverse of A.
Hint: Show that the residual $\mathbf{r} = \mathbf{b} - A\widehat{\mathbf{x}}$ *satisfies* $A^\top \mathbf{r} = \mathbf{0}$.

(b) (10 points) Now let A be arbitrary, possibly with linearly dependent columns. Recall that the Euclidean inner product and norm are defined by

$$\|\mathbf{u}\|_2 = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle} = \sqrt{\mathbf{u}^\top \mathbf{u}}.$$

The least-squares problem is

$$\min_{\mathbf{x}} \|A\mathbf{x} - \mathbf{b}\|_2.$$

Show that $\widehat{\mathbf{x}} = A^+\mathbf{b}$ is a least-squares solution. Then show that it is the unique least-squares solution of minimum Euclidean norm; that is,

$$\|\widehat{\mathbf{x}}\|_2 \leq \|\mathbf{x}\|_2$$

for every least-squares solution $\mathbf{x}$, with equality only when $\mathbf{x} = \widehat{\mathbf{x}}$.

Hint 1: Use the fact that $AA^+\mathbf{b}$ is the orthogonal projection of $\mathbf{b}$ onto $\mathrm{Col}(A)$.

Hint 2: As in Problem 1(c), write every least-squares solution as $\mathbf{x} = \widehat{\mathbf{x}} + \mathbf{z}$, where $\mathbf{z} \in \mathrm{Null}(A)$.

Remark. When a regression problem has many solutions, all of those solutions make the same predictions on the training data. The minimum-norm solution is often preferred because it uses the smallest overall parameter magnitude. It avoids adding arbitrary components from $\mathrm{Null}(A)$ that do not improve the fit. This can provide a simpler and more stable model and may improve generalization to new data.

If A is rank deficient, $A^\top A$ is not invertible, but the Moore–Penrose pseudoinverse A^+ is still defined and selects the minimum-norm least-squares solution.

Solution:

(a) By construction $A\widehat{\mathbf{x}}$ is the orthogonal projection of $\mathbf{b}$ onto $\mathrm{Col}(A)$, so the residual $\mathbf{r} = \mathbf{b} - A\widehat{\mathbf{x}}$ is orthogonal to every vector in $\mathrm{Col}(A)$. Each column $\mathbf{a}_j$ of A lies in $\mathrm{Col}(A)$, hence $\mathbf{a}_j^\top \mathbf{r} = 0$ for all j, which stacked together is

$$A^\top \mathbf{r} = A^\top(\mathbf{b} - A\widehat{\mathbf{x}}) = \mathbf{0} \quad \Longleftrightarrow \quad A^\top A\,\widehat{\mathbf{x}} = A^\top \mathbf{b}.$$

Since the columns of A are linearly independent, $A^\top A$ is invertible: Multiplying the equations by $(A^\top A)^{-1}$ gives

$$\widehat{\mathbf{x}} = (A^\top A)^{-1} A^\top \mathbf{b} = A^+\mathbf{b}.$$

(b) We will first show that $\widehat{\mathbf{x}} = A^+\mathbf{b}$ is a least-squares solution. Write $\mathbf{b}_{\text{proj}} = AA^+\mathbf{b}$, the orthogonal projection of $\mathbf{b}$ onto $\mathrm{Col}(A)$. For any $\mathbf{x}$, the vector $A\mathbf{x} - \mathbf{b}_{\text{proj}}$ lies in $\mathrm{Col}(A)$ while $\mathbf{b} - \mathbf{b}_{\text{proj}}$ is orthogonal to $\mathrm{Col}(A)$, so by the Pythagorean identity

$$\|A\mathbf{x} - \mathbf{b}\|_2^2 = \|A\mathbf{x} - \mathbf{b}_{\text{proj}}\|_2^2 + \|\mathbf{b}_{\text{proj}} - \mathbf{b}\|_2^2 \geq \|\mathbf{b}_{\text{proj}} - \mathbf{b}\|_2^2,$$

with equality if and only if $A\mathbf{x} = \mathbf{b}_{\text{proj}}$. Since $A\widehat{\mathbf{x}} = AA^{+}\mathbf{b} = \mathbf{b}_{\text{proj}}$, the vector $\widehat{\mathbf{x}}$ attains the minimum and is a least-squares solution. Moreover, $\mathbf{x}$ is a least-squares solution if and only if $A\mathbf{x} = \mathbf{b}_{\text{proj}}$.

To show $\widehat{\mathbf{x}}$ is the minimum norm solution, let $\mathbf{x}$ be any least-squares solution. Then $A(\mathbf{x} - \widehat{\mathbf{x}}) = \mathbf{b}_{\text{proj}} - \mathbf{b}_{\text{proj}} = \mathbf{0}$, so as in Problem 1(c) we may write

$$\mathbf{x} = \widehat{\mathbf{x}} + \mathbf{z}, \qquad \mathbf{z} \in \text{Null}(A).$$

The pseudoinverse satisfies $\widehat{\mathbf{x}} = A^{+}\mathbf{b} \in \text{Col}(A^{\top})$, and $\text{Col}(A^{\top}) = \text{Null}(A)^{\perp}$, so $\widehat{\mathbf{x}}^{\top}\mathbf{z} = 0$. Hence

$$\|\mathbf{x}\|_2^2 = \|\widehat{\mathbf{x}} + \mathbf{z}\|_2^2 = \|\widehat{\mathbf{x}}\|_2^2 + 2\widehat{\mathbf{x}}^{\top}\mathbf{z} + \|\mathbf{z}\|_2^2 = \|\widehat{\mathbf{x}}\|_2^2 + \|\mathbf{z}\|_2^2 \geq \|\widehat{\mathbf{x}}\|_2^2,$$

so $\|\widehat{\mathbf{x}}\|_2 \leq \|\mathbf{x}\|_2$. Equality holds if and only if $\|\mathbf{z}\|_2 = 0$, i.e. $\mathbf{z} = \mathbf{0}$ and $\mathbf{x} = \widehat{\mathbf{x}}$. Therefore $\widehat{\mathbf{x}} = A^{+}\mathbf{b}$ is the unique least-squares solution of minimum Euclidean norm.

3. Chain Rule **(25 points)**

Suppose I have a function $q : \mathbb{R}^d \to \mathbb{R}$ defined as

$$q(x) = f(g(x)), \quad x \in \mathbb{R}^d,$$

where $g : \mathbb{R}^d \to \mathbb{R}$ is defined as $g(x) = \sum_{i=1}^{d} x_i$ (namely, it takes summation over all entries of x), and $f : \mathbb{R} \to \mathbb{R}$ is defined as $f(y) = y^4$. Derive the expression of $\frac{\partial q(x)}{\partial x}$.

Solution: Write $y = g(x) = \sum_{i=1}^{d} x_i = \mathbf{1}^{\top}x$, where $\mathbf{1} = (1, \ldots, 1)^{\top} \in \mathbb{R}^d$. Then $q(x) = f(y) = y^4$.

$$f'(y) = 4y^3, \qquad \frac{\partial g(x)}{\partial x_i} = \frac{\partial}{\partial x_i}\sum_{j=1}^{d} x_j = 1 \quad \text{for every } i, \qquad \text{so} \qquad \frac{\partial g(x)}{\partial x} = \mathbf{1}.$$

By the chain rule, for each coordinate i,

$$\frac{\partial q(x)}{\partial x_i} = f'(g(x)) \cdot \frac{\partial g(x)}{\partial x_i} = 4\Big(\sum_{j=1}^{d} x_j\Big)^3 \cdot 1.$$

Collecting the d coordinates into a vector gives

$$\frac{\partial q(x)}{\partial x} = 4\Big(\sum_{i=1}^{d} x_i\Big)^3 \mathbf{1} = 4(\mathbf{1}^{\top}x)^3\,\mathbf{1} \in \mathbb{R}^d,$$

i.e. every entry of the gradient equals $4\left(\sum_{i=1}^{d} x_i\right)^3$.

4. Chain Rule for Vector-valued Function (20 points)

Suppose I have a scalar function q defined as

$$q(x; a, b) = b \cdot \phi\big(a^\top \phi\big(x\big)\big), \quad x \in \mathbb{R}^d.$$

In particular,

- The function $\phi(x) = \log(1 + e^x)$ operates componentwisely (namely, if x is a scalar, $\phi(x) = \log(1 + e^x)$ is a scalar; if x is a d-dimensional vector, $\phi(x) = (\log(1 + e^{x_1}), \ldots, \log(1 + e^{x_d}))^\top$ is also a d-dimensional vector).
- $a \in \mathbb{R}^d$ and $b \in \mathbb{R}$ are weight parameters.

Answer the following questions:

(a) (8 points) For $x \in \mathbb{R}^d$, figure out the dimensions of $\phi(x)$, $a^\top \phi(x)$, $\phi\big(a^\top \phi\big(x\big)\big)$, and $q(x; a, b)$, respectively.

(b) (12 points) Calculate the expressions of $\frac{\partial q(x;a,b)}{\partial x}$, $\frac{\partial q(x;a,b)}{\partial a}$, $\frac{\partial q(x;a,b)}{\partial b}$.

Solution:

(a)

$$\phi(x) \in \mathbb{R}^d, \qquad a^\top \phi(x) \in \mathbb{R}, \qquad \phi\big(a^\top \phi(x)\big) \in \mathbb{R}, \qquad q(x; a, b) \in \mathbb{R}.$$

That is, $\phi(x)$ is a d-dimensional vector (since ϕ acts componentwise), $a^\top \phi(x)$ is a scalar (inner product of two d-vectors), the outer ϕ applied to a scalar is again a scalar, and multiplying by $b \in \mathbb{R}$ leaves a scalar.

(b) **Notation.** The softplus $\phi(u) = \log(1 + e^u)$ has derivative

$$\phi'(u) = \frac{e^u}{1 + e^u} = \frac{1}{1 + e^{-u}} = \sigma(u),$$

the logistic sigmoid. For a vector argument, write $\phi'(x) = (\sigma(x_1), \ldots, \sigma(x_d))^\top$, again componentwise.

Introduce the intermediate quantities

$$h = \phi(x) \in \mathbb{R}^d, \qquad s = a^\top h \in \mathbb{R}, \qquad t = \phi(s) \in \mathbb{R}, \qquad q = b\, t \in \mathbb{R}.$$

Derivative with respect to b. Since q is linear in b,

$$\frac{\partial q}{\partial b} = t = \phi\big(a^\top \phi(x)\big) = \log\Big(1 + e^{\,a^\top \phi(x)}\Big) \in \mathbb{R}.$$

Derivative with respect to a. Only $s = a^\top h$ depends on a, and $\partial s/\partial a = h$. By the chain rule,

$$\frac{\partial q}{\partial a} = b\, \phi'(s)\, \frac{\partial s}{\partial a} = b\, \sigma\big(a^\top \phi(x)\big)\, \phi(x) \in \mathbb{R}^d.$$

Derivative with respect to x. Now s depends on x through $h = \phi(x)$. Since ϕ acts componentwise, the Jacobian $\partial h/\partial x = \operatorname{diag}\big(\sigma(x_1), \ldots, \sigma(x_d)\big)$ is diagonal, so

$$\frac{\partial s}{\partial x} = \Big(\frac{\partial h}{\partial x}\Big)^{\top} a = \operatorname{diag}\big(\sigma(x_1), \ldots, \sigma(x_d)\big)\, a = a \odot \phi'(x),$$

where $\odot$ denotes the componentwise (Hadamard) product. Therefore

$$\frac{\partial q}{\partial x} = b\,\phi'(s)\,\frac{\partial s}{\partial x} = b\,\sigma\big(a^{\top}\phi(x)\big)\,\big(a \odot \phi'(x)\big) \in \mathbb{R}^d,$$

or componentwise,

$$\frac{\partial q}{\partial x_i} = b\,\sigma\big(a^{\top}\phi(x)\big)\,a_i\,\sigma(x_i), \qquad i = 1, \ldots, d.$$